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WEEP-HOLE DRILLING

Abstract

Systems and methods for creating a weep hole in an uncured, concrete, hollow core structural component having one or more external surfaces and one or more internal surfaces. The weep hole is drilled using a drilling apparatus comprising a drill, a rotating drill bit with an internal fluid passageway, a power source, and a pressurized fluid source. Loose, uncured concrete material deposited near the weep hole on the one or more internal surface as a result of drilling the weep hole is removed by releasing pressurized fluid which flows from the pressurized fluid source, through the drill and the rotating drill bit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present patent application claims priority to and is a continuation of the non-provisional application identified by U.S. Ser. No. 18/519,763, filed Nov. 27, 2023, which is a continuation of the non-provisional application identified by U.S. Ser. No. 18/165,055, filed Feb. 6, 2023, which issued as U.S. Pat. No. 11,828,110 on Nov. 28, 2023, which is a continuation of the non-provisional application identified by U.S. Ser. No. 17/245,702, filed Apr. 30, 2021, which issued as U.S. Pat. No. 11,572,741 on Feb. 7, 2023, which claims priority to the provisional patent application identified by U.S. Ser. No. 63/18,085, filed on Apr. 30, 2020, the entire content of each of which is hereby incorporated herein by reference.

FIELD OF THE INVENTION
[0002] The present disclosure relates generally to creating a weep hole in a fully or partially cured concrete structural component such that material removed from the concrete structural component due to forming the weep hole does not interfere with functionality of the weep hole.
BACKGROUND OF THE INVENTION
[0003] A hollow core floor is a precast slab of concrete constructed with multiple, continuous, interior voids that run the length of the slab. These voids allow the slab to maintain its structural strength while significantly reducing its weight and material requirement. A serious concern associated with hollow core flooring is the potential for water entering and collecting inside these voids. If a significant volume of water is able to collect in a void, then the increased weight can cause additional stresses on the structural members of a building. Additionally, if a void that contains water experiences freezing temperatures, then the thermal expansion of the freezing water could cause the slabs to crack and weaken. Even if only a small amount of moisture is able to accumulate due to improper ventilation, many health problems and damage to building components can occur as a result of the growth of mold and bacteria. It is therefore extremely important to permit the escape of water from the interior voids of hollow core floor members in order to prevent severe and permanent damage to a structure.
[0004] In order to avoid these types of problems weep holes are created on the bottom-side of each void to allow water to drain from the void. The conventional method for creating weep-holes in a concrete structure is to drill a hole using a drill and a masonry drill bit. The drilling location should be positioned in-line with the center of the void while the concrete is either fully or partially cured. It is desirable to drill the weep hole while the concrete is only partially cured, because the concrete will be softer, thus requiring less physical labor and extending the tool life of the drill bit. When creating the weep hole, a head of a masonry drill bit is positioned in the desired location for the hole. Power is then applied to a drill which has the masonry drill bit attached. The head of the masonry drill bit cuts the concrete material, and the flutes of the bit lift the cut material (i.e., concrete debris) from the hole and deposit the cut material adjacent to the drilling area. Once the drill bit has passed through the concrete material and has entered into the core area, the bit is removed, and the weep hole is complete. A major problem associated with the aforementioned method for creating weep holes is that the removed concrete debris is only displaced directly adjacent to the drilling surface and oftentimes falls back into the newly drilled hole once the bit is removed. This debris fills or forms a barrier surrounding the newly drilled hole causing the weep

hole to be effectively blocked and ineffective. When the weep hole is drilled from within the void while the concrete is only partially cured, the partially cured debris may fall into the newly drilled hole before fully curing. If the partially cured debris is not promptly removed, then the material may become fully cured in the previously drilled weep hole, requiring additional weep hole drilling to remove the material. The conventional methods do not offer a satisfactory solution for creating a weep hole free from debris.

[0005] Thus, a need exists for a device and method for creating a weep hole in a concrete structural component, while ensuring that the debris resulting from the creation of the weep hole is removed from the immediate area of the newly created weep hole.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0006] Like reference numerals in the figures represent and refer to the same or similar elements of functions. Implementations of the disclosure may be better understood when consideration is given to the following detailed description thereof. Such description makes reference to the annexed pictorial illustrations, schematics, graphs, drawings and appendices. In the drawings:

[0007] FIG. **1** is a diagrammatic view of an exemplary drilling apparatus constructed in accordance with the present disclosure.

[0008] FIG. **2** is a side elevational view of an exemplary drill bit utilized in accordance with the present disclosure.

[0009] FIG. **3** is a cross-sectional view of the drill bit of FIG. **1**, taken along the lines **3**-**3**.

[0010] FIGS. **4**A-**4**D are diagrammatic, side elevation views of the drilling apparatus being utilized to form a weep hole in a hollow core structural component in accordance with the present disclosure.

[0011] FIGS. **5**A-**5**F are exemplary diagrammatic perspective views from within a void of the hollow core structural component showing steps for forming a weep hole substantially free of debris, in accordance with the present disclosure.

[0012] FIG. **6** is a logic flow diagram illustrating an exemplary method for forming a weep hole that is substantially free of debris, within a hollow core structural component in accordance with the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0013] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0014] In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the inventive concept. This description should be read to include one or more and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0015] Further, use of the term “plurality” is meant to convey “more than one” unless expressly stated to the contrary.

[0016] As used herein, qualifiers like “substantially,” “about,” “approximately,” and combinations and variations thereof, are intended to include not only the exact amount or value that they qualify, but also some slight deviations therefrom, which may be due to manufacturing tolerances, measurement error, wear and tear, stresses exerted on various parts, and combinations thereof, for example.

[0017] The use of the term “at least one” or “one or more” will be understood to include one as well as any quantity more than one. In addition, the use of the phrase “at least one of X, V, and Z” will be understood to include X alone, V alone, and Z alone, as well as any combination of X, V, and Z.

[0018] The use of ordinal number terminology (i.e., “first”, “second”, “third”, “fourth”, etc.) is solely for the purpose of differentiating between two or more items and, unless explicitly stated otherwise, is not meant to imply any sequence or order or importance to one item over another or any order of addition.

[0019] Finally, as used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0020] Referring now to the drawings, FIG. **1** illustrates a drilling apparatus **10** for creating a weep hole **12** (see FIG. **4**D) in a hollow core structural component **14**. The drilling apparatus **10** may be used for drilling the weep hole **12** and removing any material debris **16** generated during a drilling process as depicted in FIGS. **4**A-**4**D, FIGS. **5**A-**5**D, and FIGS. **6**A-**6**F. Broadly, the drilling apparatus **10** may include a drill bit **20**, connected to a drill **21**.

[0021] Referring to FIG. **1-3**, the drill bit **20** may have a shank **22**, a neck **23**, and a body **24**. The shank **22** is shown to have a hex shank, but it should be understood that the shank **22** may be a brace shank, a straight shank, a square shank, a SDS shank, a threaded shank or any other type of shank that would securely attach the drill bit **20** to the drill **21**. The neck **23** of the drill bit **20** may have a substantially cylindrical shape, although the shape of the neck **23** can vary. In one embodiment, the neck **23** has a diameter equal to or less than a diameter of the body **24**. The length of the neck **23** may vary and be dependent upon the dimension of the hollow core structural component **60**.

[0022] The body **24** may include a cutting edge **29** configured to cut and remove material from the hollow core structural component **60**, and flutes **25** that lift the material debris **16** out of the weep hole **12**. The length and diameter of the body **24** will vary and will be determined by the depth and diameter requirements for the weep hole **12**. The drill bit **20** also includes an internal fluid passageway **27** so that pressurized fluid can enter a fluid inlet **26**, travel through the internal fluid passageway **27**, and exit a fluid outlet **28**. The fluid can be a gas or liquid. For example, the fluid can be air. The fluid inlet **26** will be located in a position at or near the shank **22** so that when the shank **22** is firmly seating in a chuck **32** of the drill **21** the fluid inlet **26** will be fluidly connected to a fluid outlet within the drill **21**. The internal fluid passageway **27** may be a generally cylindrical void that extends from the fluid inlet **26** to the fluid outlet **28**. The diameter of the internal fluid passageway **27** will be determined based on ensuring that an adequate volume of pressurized fluid is able to flow through the internal fluid passageway **27** at a specified fluid pressure without jeopardizing the structural integrity of the drill bit **20**. The fluid outlet **28** may be located at any location along an external surface of the body **24** or the neck **23** of the drill bit **20** so long as the fluid outlet **28** can direct fluid around the weep hole **12** to remove the material debris **16** as described herein. It may, however, be preferable to position the fluid outlet within the body **24** so as to be closer to the cutting edge **29** to greater control the removal of the material debris **16** when releasing pressurized fluid. The size and shape of the fluid outlet **28** may be designed to direct pressurized fluid exiting the fluid outlet **28** in order to maximize the removal of material debris **16** located near the weep hole **12**.

[0023] The drill **21** includes the chuck **32** that is configured to connect to the drill bit **20**, and a motor (not shown) included within a housing **33**. In the example shown, the drill **21** is powered by electricity, although other forms of powering the drill **21** can be used. For example, the drill **21** can be a pneumatic drill powered by pressurized fluid. In some embodiments, the drill **21** is portable and adapted to be utilized by an operator to manually form the weep hole **12**. In this embodiment, the drill **21** includes a handle **34** that can be gripped by the operator. In other embodiments, the drill **21** is connected to a guide and operated as part of a larger machine to form the weep hole **12**. In some embodiments, the drilling apparatus **10** may include multiple drills **21** connected to multiple bits **20** that are guided simultaneously (or separately) to form multiple weep holes **12**. In some embodiments, the drill **21** is a hammer drill, while in other embodiments, the drill **21** is not a hammer drill.

[0024] In the example shown, the drill **21** is connected to a power source **40** that provides the energy to enable the drill **21** to rotate the drill bit **20**. The power source **40** may provide power in the form of electric, hydraulic, or pneumatic power. The drilling apparatus **10** includes a power switch **42** that controls the flow of power from the power source **40** to the drill **21**. Although FIG. **1**, depicts the power switch **42** as a trigger switch located near the handle **34**, it should be understood by a person skilled in the art that the power switch **42** does not need to be fixed to the drill **21**. The power switch **42** may be operated manually in the form of a push button switch, a toggle switch, a rotary cam switch, a valve, or any other similar device. Alternatively, the power switch may be operated autonomously by way of computer software instruction. When the power switch **42** is switched on, power from the power source **40** is transmitted to the drill **21**, causing the drill **21** to rotate the chuck **32** and the drill bit **20** in a desired direction. The power switch **42** may allow regulation of the power being supplied to the drill **21** from the power source **40** to provide control of a rotational speed and a torque for the drill **21**. Alternatively, the rotational speed and torque of the drill may be controlled by other mechanisms.

[0025] The drill **21** is also connected to a pressurized fluid source **50** that provides pressurized fluid to the fluid outlet **28** of the drill bit **20** via the drill **21**. The pressurized fluid source **50** may be a pressurized fluid tank or a fluid compressor. The pressurized fluid source **50** may include a pressure regulator capable of providing adjustability of the pressure contained within or released from the pressurized fluid source **50**. The pressurized fluid source **50** may be connected to the drill **21** via a hose **51**. The drilling apparatus **10** also includes a pressurized fluid actuator **52** that may allow pressurized fluid to flow from the pressurized fluid source **50**, to the fluid inlet **26**, through the internal fluid passageway **27**, and exit the fluid outlet **28**. The pressurized fluid actuator **52** may be actuated by electric, hydraulic, pneumatic, or human power. For example, the pressurized fluid actuator may be implemented as a valve that is controlled via a trigger that is placed adjacent to the handle **34**, as shown in FIG. **1**. When the pressurized fluid actuator **52** is actuated, pressurized fluid from the pressurized fluid source **50** is released into the air inlet **26**. The drill **21** may have a regulator (not shown) at some location between the pressurized fluid source **50** and the fluid inlet **26** to allow for the control of the fluid pressure that exits the fluid outlet **28**.

[0026] Turning now to FIGS. **4**A-**4**D, an exemplary method of use of the drilling apparatus **10** will be described. As shown in FIG. **4**A, first the drilling apparatus **10** may be positioned so that the drill bit **20** is situated in a direction for drilling with the body **24** of the drill bit **20** at a location for drilling the weep hole **12** in a hollow core structural component **14**. The hollow core structural component **14** may have one or more external surfaces **62** and one or more internal surfaces **64** forming a cavity **66** in the hollow core structural component **14**. In one embodiment, as shown in FIGS. **4**A-**4**D, the hollow core structural component **14** may have a first one or more external surface **62***a* and a second one or more external surface **62***b*. However, it will be understood that the hollow core structural component **60** may have more or fewer external surfaces **62**.

[0027] The operator may engage the power switch **42** to provide power to the drill **21** from the power source **40** so that the drill **21** rotates the drill bit **20** in a desired rotational direction **70** and at a rotational speed for drilling the weep hole **12** through the hollow core structural component **14**. A first directional force **72***a* may be applied to the drill **21** in a desired direction for drilling the weep hole **12**.

[0028] As illustrated in FIG. **4**B, the first directional force **72***a* may be applied to drive the drill bit **20** to drill from a first one or more external surface **62***a*, through the material of the hollow core structural component **14**, through a first one or more internal surface **64***a*, and into the cavity **66**. The drilling process may cause material debris **16** from a first drilled hole **65** to be deposited just below the first drilled hole **65** and in a general location of where the weep hole **12** is to be created.

[0029] As depicted in FIG. **4**C., the first directional force **72***a* may be applied to drive the drill **21** through the cavity **66**, and allow the drill bit **20** to drill through a second one or more internal surface **64***b*, through the material of the hollow core structural component **14**, and through a second one or more external surface **62***b*. After the drill bit **20** drills through the second one or more external surface **62***b*, the fluid outlet **28** may be positioned so that the fluid outlet **28** is proximate to the second one or more internal surface **64***b*. The pressurized fluid actuator **52** may be actuated, causing pressurized fluid to flow from the pressurized fluid source **50**, through the hose **51**, the internal fluid passageway **27**, and exit the fluid outlet **28** while the drill bit **20** continues to rotate in the desired rotational direction **70**. The pressurized fluid actuator **52** may allow the pressurized fluid to continue to flow through the drill bit **20**, and the drill **21** of the drilling apparatus **10** and out the fluid outlet **28** until all loose, uncured concrete, material debris **16** is substantially removed from the immediate area surrounding the weep hole **12**. The pressurized fluid actuator **52** should allow pressurized fluid flow while the drill bit **20** continues to rotate one or more revolutions to ensure the pressurized fluid is directed at the material debris **16** about a parameter of the weep hole **12**. A second directional force **72***b* may be applied to the drill **21** so that the fluid outlet **28** is moved in and out of the weep hole **12**. The second directional force **72***b* may add a third dimension to the position of the fluid outlet **28** which may assist with the removal of material debris **16** proximate to the weep hole **68**.

[0030] As shown in FIG. **4**D, the pressurized fluid actuator **52** may be deactivated, stopping the flow of pressurized fluid through the drilling apparatus **10**. The power switch **42** may also be turned off, cutting off the power to the drill **21**, and stopping the rotation of the drill bit **20**. A third directional force **72***c* may be applied to the drilling apparatus **10** until the drill bit **20** is entirely removed from the hollow core structural component **14**. The result of the process is the creation of the weep hole **12** in the hollow core structural component **14** free of any material debris **16** that might interfere with the function of the weep hole **12**. When the hollow core structural component **14** is not fully cured when the weep hole **12** is formed, then the hollow structural component **14** may be allowed to cure prior to installation as a structural component in a structure, such as a building, road, or the like.

[0031] FIGS. **5**A-**5**F, illustrates several perspective views from within the cavity **66** of the hollow core structural component **14**, showing steps of an exemplary method for forming the weep hole **12** substantially free of material debris **16**, in accordance with the present disclosure. FIG. **5**A shows the cavity **66** of the hollow core structural component **14** before the method has been initiated toward the creation of the weep hole **68**. There may be the presence of loose, material debris **16** within the cavity **66** along the lower portion of the one or more internal surfaces **64** prior to the drilling process as a result of forming the hollow core structural component **14**.

[0032] As shown in FIG. **5**B, a force may be applied to the drill **21** in the direction of the weep hole **12** causing the drill bit **20** to drill through the first one or more external surface **62** and the first one or more internal surface **64**. The drill bit **20** continues to rotate inside the cavity **66** after the weep hole **12** has been formed. Additional material debris **16** may collect beneath the first drilled hole **65** as a result of the drilling process.

[0033] As depicted in FIG. **5**C, the force may continue to be applied to the drill **21** in the direction of the weep hole **12** causing the drill bit **20** to drill through the second one or more internal surface **64** and a second one or more external surface **62**. Additional material debris **16** may accumulate around the weep hole **12** as a result of the drilling process.

[0034] As shown in FIG. 5D, the drill bit **20** may be positioned so that the fluid outlet **28** is proximate to the second one or more internal surface **64**. The pressurized fluid actuator **52** may be actuated, allowing pressurized air to flow through the internal fluid passageway **27** while the drill **21** continues to rotate the drill bit **20**. Pressurized fluid exits the fluid outlet **28** with enough force to reposition any loose, material debris **16** a sufficient distance from the weep hole **12**. The fluid outlet **28** may be directed generally perpendicular to the drill bit **20** and designed to focus the pressurized fluid exiting the fluid outlet **28** away from the weep hole **12**. The drill bit **20** may be rotated one or more revolutions while pressurized fluid is being release from the fluid outlet **28** to ensure all material debris **16** around the weep hole **12** is sufficiently removed from the immediate area. The drill bit **20** may be moved in and out of the weep hole **12** to assist the removal of material debris **16**. By moving the drill bit **20** in and out of the weep hole **12**, the pressurized fluid exiting the fluid outlet **28** may exert a variety of forces on the material debris **16** that may not be experienced by a stationary drill bit **20**.

[0035] As shown FIG. 5E, the pressurized fluid actuator **52** may be deactivated so that pressurized fluid is no longer flowing through the drill bit **20** and the drill **21** of the drilling apparatus **10**. The power switch **42** may also be turned off, causing the drill **21** to stop rotating the drill bit **20**. A force may be applied to the drill **21** opposite of the drilling direction until the drill bit **20** is removed from the hollow core structural component **14**.

[0036] FIG. 5F, depicts the cavity **66** of the hollow core structural component **14** once the drill bit **20** has been removed. As illustrated in FIG. 5F, the weep hole **12** is free from any loose material debris **16** that might interfere with the function of the weep hole **12**.

[0037] Referring now to FIG. **6**, a drilling method **100** provides an exemplary method for forming the weep hole **12** that is substantially free of material debris **16**, within the hollow core structural component **14** in accordance with the present disclosure. In step **102** of the drilling method **100**, an operator may identify a desired location to form the weep hole **12** in the cavity **66** of the hollow core structural member **14**. The ideal location for the weep hole **12** would be the lowest area of the one or more internal surfaces **64**, but may be any location within the hollow core structural component **14** that would allow moisture to be removed from the cavity **66**. The desired location for the weep hole **12** may only be accessible by first drilling through the opposite side of the hollow core structural component **14**.

[0038] In step **104** of the drilling method **100**, the cutting edge **29** of the drill bit **20** may be placed at the desired location for the weep hole **12**. The drill **21** may need to be positioned so that the drill bit **20** will be driven in the desired direction for the weep hole **12**. Next, in step **106** of the drilling method **100**, the drill **21** may be activated with power to cause the drill bit **20** to rotate, and pressure may be applied to the drill **21** to allow drill bit **20** to bore the weep hole **12** in the hollow core structural member **14**. The drill **21** will be activated by turning on the power switch **42** which allows power to be provided to the drill **21** from the power source **40**. In step **108**, pressure may continue to be applied to the drill **21** until the drill bit **20** has bored the weep hole **12**. The amount of pressure required may depend on the characteristics of material of the hollow core structural component **14**, the rotational speed of the drill bit **20**, and the characteristics of the drill bit **20**. Depending on the location of the weep hole **12**, the drill bit **20** may need to bore a hole in one or more layers of the hollow core structural component **14** before boring the weep hole **12**.

[0039] As detailed in Step **110**, power may be maintained to the drill **21**, while the drill bit **20** may be repositioned so that the fluid outlet **28** is proximate to the one or more internal surfaces **64**, and the pressurized fluid actuator **52** is activated to release pressurized fluid form the pressurized fluid source **50**. In step **112**, power may be maintained to the drill **21** and releasing the pressurized fluid, while the drill bit **20** is repositioned in and out of the weep hole **12** until all material debris **16** has been removed from the immediate area. By performing this action, pressurized fluid will be released in a 360-degree spray about the drill bit **20** and at various points vertically along a drilling axis within the range of the in and out motion. The pressurized fluid will apply forces on the material debris **16** immediately surrounding the weep hole **12**. These forces may cause the material debris **16** to be relocated substantially away from the weep hole **12**. In one embodiment, the pressurized fluid actuator **52** should be engaged continuously for at least one revolution of the drill bit **20** to ensure pressurized fluid is released in all directions about the drilling axis.

[0040] Lastly, in step **114** of the drilling method **100**, Once the area immediately around the weep hole **12** has been clear of any loose, material debris **16**, power may be shut off to the drill **21**, the pressurized fluid actuator **52** may be deactivated to stop pressurized fluid from flowing from the pressurized fluid source **50**, and the drill bit **20** may be removed from the weep hole **12**.

[0041] While the present disclosure has been described in connection with certain embodiments so that aspects thereof may be more fully understood and appreciated, it is not intended that the present disclosure be limited to these particular embodiments. On the contrary, it is intended that all alternatives, modifications and equivalents are included within the scope of the present disclosure. Thus the examples described above, which include particular embodiments, will serve to illustrate the practice of the present disclosure, it being understood that the particulars shown are by way of example and for purposes of illustrative discussion of particular embodiments only and are presented in the cause of providing what is believed to be the most useful and readily understood description of procedures as well as of the principles and conceptual aspects of the presently disclosed methods and compositions. Changes may be made in the structures of the various components described herein, or the methods described herein without departing from the spirit and scope of the present disclosure.

Claims

1.-**14**. (canceled)

15. A system, comprising: one or more drills; two or more rotatable drill bits connected to the one or more drills, each of the two or more rotatable drill bits having an internal fluid passageway and a fluid outlet, each of the rotatable drill bits configured to drill a corresponding weep hole in an uncured, concrete, hollow core structural component having one or more external surfaces and an internal surface, resulting in two or more weep holes in the uncured, concrete, hollow core structural component, whereby loose, uncured concrete material is deposited adjacent to the two or more weep holes on the internal surface of the uncured, concrete, hollow core structural component; and one or more pressurized fluid source fluidly connected to the internal fluid passageways of the two or more rotatable drill bits, wherein when the two or more rotatable drill bits are repositioned, a flow of pressurized fluid is directed from the pressurized fluid source through the internal fluid passageway out of the fluid outlet of each of the two or more rotatable drill bits, thereby removing the loose, uncured concrete material deposited adjacent to the two or more weep holes on the internal surface of the uncured, concrete, hollow core structural component.

16. The system of claim 15, wherein the pressurized fluid is air.

17. The system of claim 15, wherein the pressurized fluid is a gas.

18. The system of claim 15, wherein the pressurized fluid is a liquid.

19. The system of claim 15, comprising: a valve fluidly positioned between the pressurized fluid source and the internal fluid passageways of the two or more rotatable drill bits, the valve having an actuated state in which the pressurized fluid in the pressurized fluid source is directed into the internal fluid passageways of the two or more rotatable drill bits, and the valve having a deactivated state in which a flow of pressurized fluid from the pressurized fluid source is stopped.

20. The system of claim 15, wherein the system comprises: a pressure regulator configured to adjust pressure within the pressurized fluid source.

21. A method, comprising: drilling, using a drilling apparatus, two or more weep holes in an uncured, concrete, hollow core structural component having one or more external surfaces and an internal surface, such that, during the drilling of the two or more weep holes, loose, uncured concrete material is generated and deposited adjacent to the two or more weep holes on the internal surface of the uncured, concrete, hollow core structural component, the drilling apparatus comprising: one or more drills; two or more rotatable drill bits connected to the one or more drills, each of the two or more rotatable drill bits having an internal fluid passageway and a fluid outlet; and one or more pressurized fluid source fluidly connected to the internal fluid passageways of the two or more rotatable drill bits; repositioning the two or more rotatable drill bits so that the fluid outlets are proximate to the internal surface of the uncured, concrete, hollow core structural component; and directing a flow of pressurized fluid from the pressurized fluid source, through the drill and the rotating drill bits, and through the fluid outlets, thereby removing the loose, uncured concrete material from adjacent to the two or more weep holes.

22. The method of claim 21, wherein the pressurized fluid is air.

23. The method of claim 21, wherein the pressurized fluid is a gas.

24. The method of claim 21, wherein the pressurized fluid is a liquid.

25. The method of claim 21, wherein the drilling apparatus comprises: a valve fluidly positioned between the pressurized fluid source and the internal fluid passageways of the two or more rotatable drill bits, the valve having an actuated state in which the pressurized fluid in the pressurized fluid source is directed into the internal fluid passageways of the two or more rotatable drill bits, and the valve having a deactivated state in which a flow of pressurized fluid from the pressurized fluid source is stopped.

26. The method of claim 21, wherein the drilling apparatus comprises: a pressure regulator configured to adjust pressure within the pressurized fluid source.